Peer-to-Peer Networking and Applications (2025) 18:311
https://doi.org/10.1007/s12083-025-02143-0

RESEARCH

BT-CSRS: A decentralized and distributed solution for sustainable seafood supply chain system utilizing zero-knowledge proofs and permissionless blockchain

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Received: 26 January 2025 / Accepted: 7 October 2025
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Abstract

The seafood supply chain faces several challenges, including lack of transparency, difficulty in traceability, and vulnerability to fraud and mislabeling. It often involves multiple intermediaries across dispersed locations, making it prone to delays, inefficiencies, and loss of product quality. Additionally, inadequate cold chain infrastructure, regulatory non-compliance, and environmental concerns further complicate the process, impacting sustainability and consumer trust. To address these challenges, this work introduces a Blockchain-based Sustainable Seafood Reacquisition Framework (BT-CSRS) that leverages Zero-Knowledge Proofs (ZKPs) and a permissionless blockchain. ZKPs are employed to validate sensitive information—such as origin, handling procedures, seafood identification, species, capture location, and adherence to sustainability standards—without disclosing the actual data. Additionally, the proposed framework integrates the seafood catching process using ZKPs features, ensuring that every stage is verifiable and trustworthy. This approach enhances transparency, mitigates fraud, and preserves the integrity of the seafood from ocean to consumer. This work employs remix IDE for implementing the proposed BT-CSRS framework on the ethereum blockchain, using solidity for developing Smart Contracts (SCs). MetaMask is utilized as the blockchain wallet for stakeholders, while the sepolia test network is used to deploy and evaluate the stability and performance of the SCs within the BT-CSRS framework. We used the JMeter simulation toolkit to perform a comprehensive set of experiments evaluating the performance of the proposed BT-CSRS framework in terms of throughput, latency, response time, and Standard (STD) deviation of response time. The proposed framework achieved an average throughput of 682.36 TPS, latency of 64.1 milliseconds, response time of 75.5 milliseconds, and a standard deviation of 68.7 milliseconds. Additionally, we summarized the deployment analysis results with detailed metrics, including SCs data, transaction hashes, block sizes, transaction indices, nonce, epochs, and other relevant parameters. The proposed BT-CSRS framework improves transparency and trust by providing secure verifiable records of seafood transactions while preserving confidentiality. This work also promotes sustainable practices by encouraging ethical sourcing through reliable data on product origin and environmental impact. Our observation reveals that our proposed framework has better features than other state-of-the-art works.

Keywords Confidential seafood reacquisition system · Blockchain · Zero-Knowledge proofs · Responsible sourcing · Environmental sustainability · Smart contract

1 Introduction

The world's oceans, covering over 70% of the earth's surface, play a critical role in sustaining life, regulating the climate, and supporting global food security [1–3]. Seafood, in particular, provides a substantial portion of the world's protein intake and supports the livelihoods of millions [4–6]. However, the integrity and reliability of the seafood industry are increasingly under threat due to challenges such as mislabelling, fraud, and irresponsible harvesting practices [7–9]. These issues not only undermine consumer trust but also lead to significant economic losses for legitimate producers and contribute to ecological degradation [10–12]. The seafood supply chain, which spans multiple stakeholders and

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Published online: 24 October 2025

geographies, suffers from a lack of end-to-end visibility, traceability, and accountability [13, 14]. Despite the use of digital tracking systems and centralized databases, current solutions frequently fall short in guaranteeing data integrity, preventing manipulation, and safeguarding confidential information [15, 16]. These systems are limited by their centralized nature, lack transparency, and are often unable to scale securely across global networks [17–19].

Blockchain Technology (BT) has emerged as a transformative solution in addressing these limitations [7, 20–22]. Its decentralized and immutable architecture enables secure, tamper-evident recording of transactions across a distributed network [23–25]. In the context of the seafood supply chain, blockchain can facilitate transparent traceability by logging the journey of seafood from the point of harvest to the consumer's plate [13, 26, 27]. This enhances accountability and can deter fraudulent practices. However, while blockchain ensures transparency and data integrity, it lacks inherent mechanisms to safeguard sensitive or proprietary information [11, 28, 29]. For industries like seafood—where harvesting locations, supplier identities, and trade secrets need to remain confidential—this presents a significant limitation [30, 31].

To address this, ZKPs offer a crucial privacy-preserving complement to blockchain. ZKPs are advanced cryptographic techniques that allow one party to prove the truth of a statement without revealing the underlying data [32, 33]. In seafood supply chains, ZKPs can be used to verify critical claims—such as compliance with regulations or quality verification—without exposing sensitive operational details [34, 35]. This enables stakeholders to maintain confidentiality while ensuring that the data remains verifiable. ZKPs have already demonstrated utility in privacy-sensitive domains such as finance and healthcare, where trust and privacy are paramount [36, 37]. Their integration into supply chains can significantly enhance both trust and data protection. When combined, blockchain and ZKPs offer a powerful synergy: blockchain provides immutable transparency, while ZKPs enforce selective disclosure, thereby ensuring both trust and privacy [38, 39].

Through this work, the potential to significantly improve the reliability and integrity of the seafood supply chain can be realized by combining BT with ZKPs. This integration aims to address persistent challenges such as data tampering, lack of transparency, weak authentication mechanisms, and the inability to securely manage product returns or reacquisitions. In particular, seafood reacquisition—referring to the return or validation of previously sold seafood products—poses challenges related to authenticity, contamination risk, and traceability. This study aims to (i) evaluate the limitations of existing centralized and semi-digital seafood tracking and return systems, (ii) demonstrate how blockchain outperforms traditional solutions in maintaining verifiable traceability and data integrity, (iii) investigate how ZKPs can ensure privacy-preserving validation of product information, and (iv) examine how these technologies can support secure and trusted seafood reacquisition processes. The broader objective is to contribute to the development of a transparent, privacy-aware, and ethically responsible seafood supply chain that promotes trust among consumers, supports legitimate stakeholders, and enables more reliable handling of product returns.

We have organized this paper as follows: In Section 2, we conduct an analysis of relevant literature to provide context for our proposed approach. Section 3 elaborates on the proposed BT-CSRS framework. Section 4 presents the implementation details, followed by analysis of the results and simulation outcomes. Section 5 discusses the comparative evaluation, real-world deployment challenges, and summarizes the key insights obtained from the framework. Finally, Section 6 concludes the paper.

2 Related work

Several recent studies have explored the integration of BT with privacy-preserving mechanisms and digital requisition systems to enhance supply chain transparency, traceability, and operational efficiency.

Sahai et al. [32] introduced a blockchain-based supply chain paradigm that seamlessly integrates privacy, traceability, and contamination tracking. The study emphasizes the use of ZKPs to ensure product integrity and data confidentiality without revealing the complete transaction history. The approach was validated on Hyperledger Fabric, demonstrating the practical efficiency of ZKPs in enhancing privacy and transparency within complex supply chain environments.

Similarly, Bai et al. [40] presented a conceptual framework distinguishing between fully decentralized ("blockchain-complete") and partially decentralized ("blockchain-partial") systems. By proposing a "proof-chain" model and leveraging ZKPs, the study provides key insights into privacy-strength trade-offs in hybrid blockchain environments. Sedlmeir et al. [41] extended this concept by proposing succinct ZKPs to balance scalability with privacy, addressing challenges in capturing fine-grained energy data in green energy systems.

Abadi et al. [42] contributed a formal cryptographic foundation by integrating timestamped primitives within the Universal Composability (UC) framework. Their approach supports UC-secure implementations of signatures and ZKPs over a global ledger, effectively mitigating backdating and postdating attacks in blockchain-based systems.

In the context of construction supply chains, Xiong et al. [43] developed a blockchain-based framework focusing on payment security and private-key protection. Their system integrates a key distribution protocol using secret sharing and network security schemes, experimentally validating the robustness and feasibility of their approach. Similarly, Malik et al. [44] proposed TradeChain, a dual-ledger supply chain framework that employs Self-Sovereign Identity (SSI), ZKPs, and Ciphertext Policy Attribute-Based Encryption (CP-ABE) to enforce dynamic access control while ensuring privacy and traceability. Implemented using Hyperledger Indy and Fabric, TradeChain demonstrates minimal performance overhead while enhancing data accountability.

A growing body of work has also addressed requisition systems in construction, healthcare, and corporate environments. Adedeji et al. [45] developed a Web-Based Material Requisition System (W-BMRS) using the MVC architecture to streamline on-site material requests and integrate ICT in construction workflows. Nadhiroh et al. [46] proposed a RAD-based system for non-medical goods requisition in hospitals, addressing inefficiencies related to delays, inventory errors, and documentation gaps. Singh et al. [47] further analyzed requisition and purchase order workflows, highlighting the impact of approval delays and data inconsistencies on inventory and procurement performance.

Other requisition-specific innovations include Reqscan by Bach et al. [48], a lightweight tool for digitizing barcoded lab requisitions, and the Financial Requisition System (FRS) proposed by Cole et al. [49], which applies Agile principles to modernize and streamline financial requisition workflows.

Enterprise-oriented applications of blockchain for procurement were demonstrated by Wang et al. [50], where Hyperledger Fabric is utilized to manage requisition, warehousing, and payment activities within internal value chains. The system demonstrated improved transparency and operational control, although the complexity of integration with legacy systems was acknowledged as a limiting factor.

Finally, Islam et al. [51] proposed a blockchain-based IoT data acquisition framework integrating UAVs and Mobile Edge Computing (MEC) for secure transmission and consensus-based storage. While the system addresses data integrity and privacy, it faces scalability and energy limitations. In the public sector, Arendt et al. [52] advocated for smart contract-based automation of low-risk government requisitions, aiming to reduce manual overheads and enhance transparency, albeit facing regulatory and integration challenges.

Despite progress in blockchain-enabled supply chain and requisition systems, current literature exhibits several limitations in addressing the specific challenges of the seafood sector—particularly in the context of reacquisition processes. While some studies explore requisition workflows without blockchain integration, others focus solely on traceability using permissioned or hybrid platforms, which often lack transparency and decentralization. Furthermore, privacy-preserving mechanisms such as ZKPs are rarely applied in conjunction with SCs to manage sensitive supply chain events, such as product returns or reacquisitions. To bridge these gaps, this research introduces the BT-CSRS, a novel ethereum-based framework that securely manages the validation of returned seafood items. By combining SCs for automated and auditable transaction control with ZKPs for confidential data verification, the proposed system enhances trust, supports responsible sourcing, and contributes to the overall traceability and sustainability of seafood supply chain operations.

3 Proposed seafood reacquisition framework

In this section, we present our proposed framework BT-CSRS by utilizing ZKPs and the ethereum blockchain. The implementation leverages the sepolia testnet for development and testing purposes. Ethereum offers a well-established infrastructure, high community support, and compatibility with SCs standards, making it suitable for implementing transparent and decentralized supply chain systems. To ensure privacy and verifiability of sensitive requisition details, the model integrates Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (zk-SNARKs) as the ZKPs protocol. The zk-SNARKs protocol enables the generation of compact, non-interactive proofs with fast verification times, aligning with the confidentiality constraints of supply chain environments that require both privacy assurance and computational efficiency. This combination enables participants to prove the validity of requisition requests without revealing confidential information such as product quantity, origin, or supplier identity.

The proposed BT-CSRS framework, illustrated in Fig. 1, integrates BT and ZKPs to ensure end-to-end traceability, data privacy, and automated verification within the supply chain. The process begins at the aquatic production stage, where ZKPs of Catching are generated to validate legally compliant and environmentally sustainable harvesting without revealing sensitive operational data. As seafood moves through the storage, transportation, and distribution phases, secure data exchange mechanisms are employed to record critical parameters such as

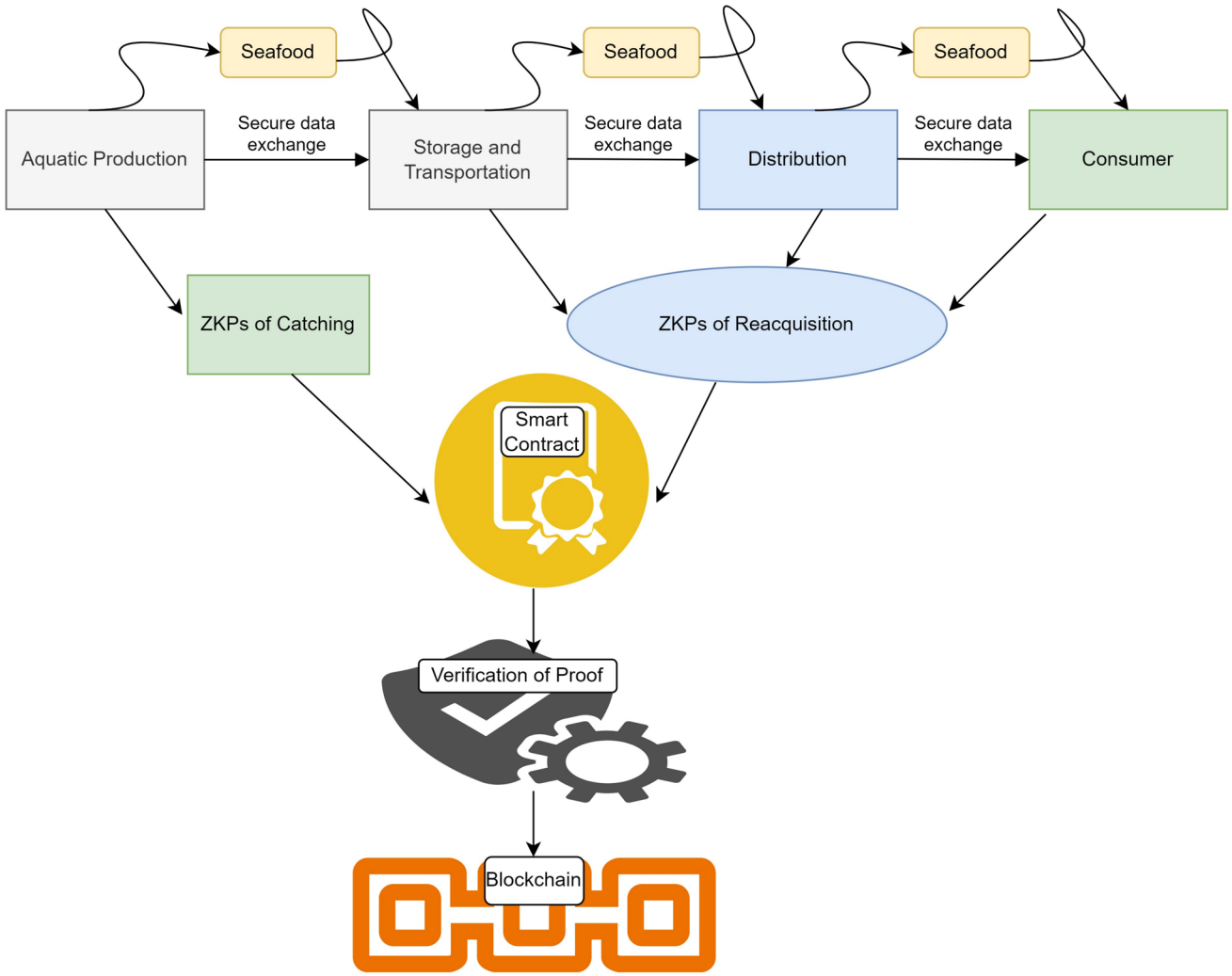


Fig. 1 Proposed BT-CSRS Seafood Reacquisition Framework

temperature and handling conditions. These records are essential for validating product quality; however, their confidentiality is preserved using ZKPs, which prevent the exposure of sensitive operational data. When a consumer initiates a return request, typically due to quality concerns, the system triggers the reacquisition ZKPs, which validate whether the seafood is maintained under acceptable cold-chain conditions and packaging standards without disclosing the exact values. These proofs are processed by a SCs, which automatically evaluates the legitimacy of the return request based on predefined rules. The SCs then initiates a verification of proof process, ensuring the integrity of the ZKPs. Once validated, the transaction is immutably recorded on the blockchain, maintaining a transparent and tamper-resistant ledger of seafood movement and returns.

Additionally, the proposed framework ensures compatibility with regulatory compliance requirements by providing transparent, verifiable, and immutable records of seafood handling and reacquisition processes. The use of blockchain guarantees that each event, such as catch, transportation, and return, is securely time-stamped and cryptographically linked, enabling trusted traceability throughout the supply chain. SCs enforce predefined compliance rules automatically, ensuring that only valid and authorized transactions are recorded. Additionally, the integration of zk-SNARKs preserves the confidentiality of sensitive information while still allowing for verifiable compliance checks.

Furthermore, the framework emphasizes the role of serial number authentication within the returned seafood workflow, leveraging confidential ZKPs to verify

unique identifiers while ensuring data privacy. This cryptographic mechanism ensures secure and trustworthy verification, safeguarding against unauthorized access and mitigating the risk of fraudulent returns. To further validate the authenticity and condition of returned items, the framework integrates multiple evaluation parameters, including quality inspection, temperature history tracking, and packaging condition assessment. By preserving data confidentiality while enabling reliable validation, the system provides a structured and secure approach for distributors to manage returns effectively, ensuring operational integrity and regulatory compliance.

The integration of ethereum with zk-SNARKs ensures a balanced approach to transparency and data confidentiality, both essential for real-world seafood supply chain operations. Blockchain provides secure time-stamped cryptographically linked records of key events such as capture, transportation, and return, thereby maintaining traceability and data integrity. SCs enforce compliance automatically, while zk-SNARKs enable verification of sensitive information without revealing actual data. This framework supports real-time auditing, enhances data accountability, and aligns with practical industry workflows. It allows seamless integration into existing systems, strengthens regulatory compliance, and minimizes the risk of violations.

To demonstrate the practical application of the proposed BT-CSRS framework, consider a real-life scenario within the seafood requisition system. When a retailer initiates a return due to concerns over freshness, the blockchain-based SCs automatically verifies key parameters such as the product's serial number, packaging condition, and temperature history, previously recorded immutably on the blockchain. To preserve privacy, ZKPs confirm that the seafood is transported under acceptable cold-chain conditions without disclosing actual temperature values. If the return conditions satisfy the predefined threshold criteria, the SCs authorizes the return and updates the ledger. This example illustrates how the integration of SCs with ZKPs can streamline the return process, ensure compliance, and protect sensitive logistical data in real-world operations.

However, to ensure secure, verifiable, and privacy-preserving seafood requisition, the proposed framework integrates ethereum-based SCs and ZKPs to manage key lifecycle events, including catching, transportation, distribution, and return. As illustrated in Algorithm 1, the system begins with initialization by registering stakeholders and deploying verification logic via SCs. ZKPs are employed to verify seafood legality and sustainability compliance without revealing sensitive data, ensuring confidentiality during the catching and reacquisition phases. Throughout the supply chain, encrypted records of transportation and storage conditions are logged on-chain to maintain end-to-end traceability. SCs enforce reacquisition rules such as quality and temperature thresholds, and only validated events are committed to the blockchain. The framework also incorporates fault detection and recovery mechanisms, ensuring data integrity and system reliability under varying conditions.

Algorithm 1 BT-CSRS Framework for Sustainable Seafood Requisition.

Require: Input parameters: $\mathcal{D}_{\text{seafood}} = \{\text{catch, transport, distribute, return}\}$, ZKPs, SCs, blockchain ledger $\mathcal{B}$

Ensure: $\mathcal{B}$ maintains validated records of traceability and reacquisition processes

Step 1: System Initialization

Define the set of entities $\mathcal{E} = \{P, D, C\}$ where P: Producers, D: Distributors, C: Consumers

Establish secure communication channels $\mathcal{C}_{\text{secure}}$ between all $e_i \in \mathcal{E}$

Deploy SCs on Ethereum to enforce verification logic $\mathcal{V}(\cdot)$

Step 2: Catch Verification via ZKPs

For each catch event c_i, check regulatory compliance: $\mathcal{R}(c_i) = 1$

Evaluate sustainability metrics $\mathcal{S}(c_i)$ (e.g., legal methods, species quota)

Generate ZKPs π_{catch} for constraint system $\mathcal{C}_{\text{catch}}$:

$$\mathcal{C}_{\text{catch}}(c_i) = 1 \iff \text{Valid Catch}$$

Submit $(\pi_{\text{catch}}, h(c_i))$ to blockchain via SCs: $\mathcal{B} \leftarrow \text{SCs}(c_i, \pi_{\text{catch}})$

Step 3: Secure Transport Data Logging

For each transport event t_j, encrypt data $\text{Enc}(t_j)$ and log metadata:

$$\mathcal{B} \leftarrow \{t_j.\text{time}, t_j.\text{condition}, h(t_j)\}$$

Step 4: Distribution and Reacquisition via ZKPs

Distribute products to endpoints $e_k \in \{D, C\}$

if Return request r_i is issued **then**

    Evaluate ZKPs π_{return} against constraint $\mathcal{C}_{\text{return}}(r_i) = 1$

    Verify parameters: $T(r_i)$ = temperature log, $Q(r_i)$ = quality score, $P(r_i)$ = packaging integrity

end if

Validate item serial identifier SID_i using ZKPs: π_{sid}

Append $(\pi_{\text{return}}, h(r_i))$ to blockchain: $\mathcal{B} \leftarrow \text{SCs}(r_i, \pi_{\text{return}})$

Step 5: Enforcement via SCs

Define threshold function Θ: $\Theta(Q(r_i)) \geq \delta$

if $\mathcal{V}(\pi_{\text{return}}) = 0 \vee \Theta(Q(r_i)) < \delta$ **then**

    Trigger exception and proceed to Step 7

else

    Auto-validate and log reacquisition via SCs

end if

Step 6: Blockchain Update

For all lifecycle events $e \in \mathcal{D}_{\text{seafood}}$, record $(h(e), \pi_e)$ on ledger $\mathcal{B}$

if All verifications $\mathcal{V}(\pi_e) = 1$ **then**

    Append block $\mathcal{B}_{\text{new}}$ to chain: $\mathcal{B} \leftarrow \mathcal{B} \cup \mathcal{B}_{\text{new}}$

end if

Step 7: Fault Recovery

Implement error function $\mathcal{E}_{\text{detect}}()$ and retry mechanism $\mathcal{R}_{\text{retry}}()$

Notify relevant $e_i \in \mathcal{E}$ of failure state

Re-execute $\mathcal{V}(\pi)$ upon resolution of constraint violation

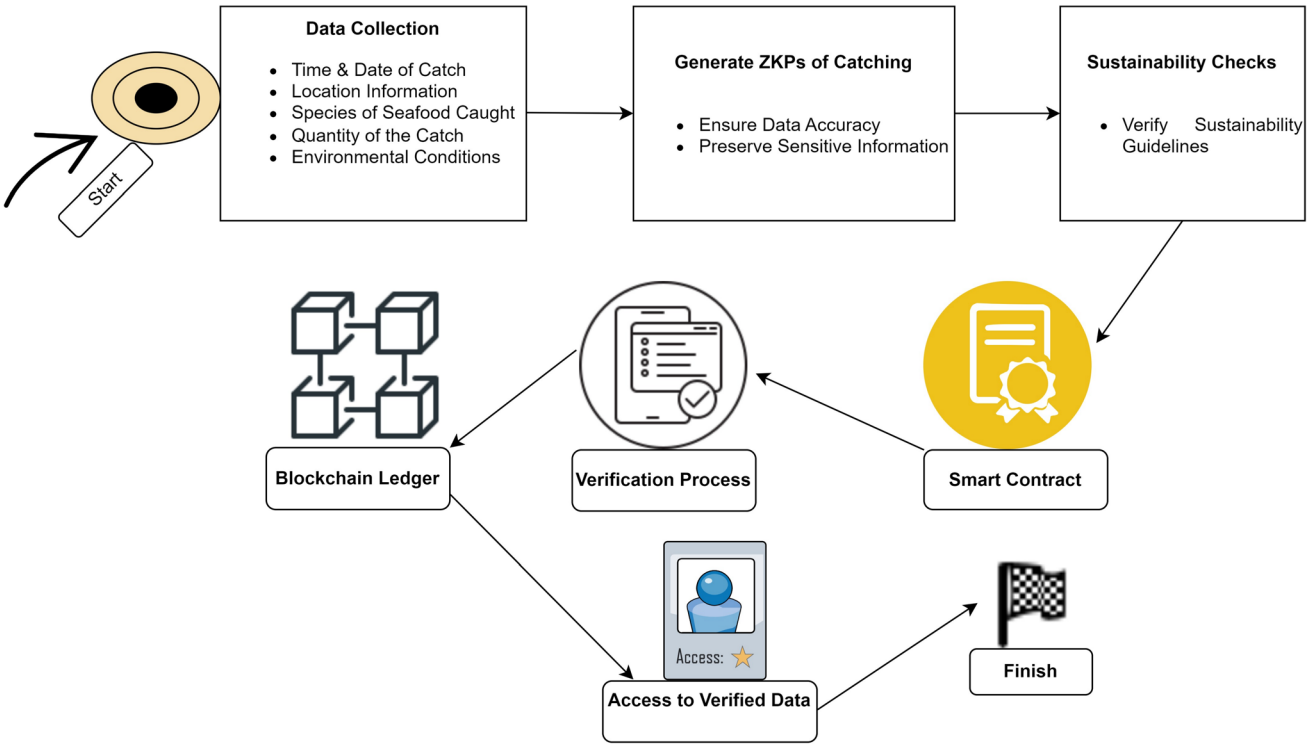


Fig. 2 ZKP-Based Seafood-Catching process

3.1 ZKPs based seafood catching process

In this subsection, we present the detailed process of the ZKPs based seafood-catching mechanism. As shown in Fig. 2, the ZKPs module for seafood catching is a key part of the proposed framework, developed to ensure the legality and environmental compliance of seafood harvesting without revealing sensitive information. To explain this process clearly, it can be broken down into several main steps.

The first step involves data collection at the time of harvest. When seafood is caught, important information is recorded, including the date and time of the catch, location details, type of species, quantity, required permits or licenses, and environmental factors such as water temperature and depth. This information is used in the next stage to generate a ZKPs, allowing the system to confirm the validity of the catch while keeping all sensitive details private.

After data collection, the next important phase involves generating ZKPs for seafood-catching activity. This step uses cryptographic methods, including protocols such as zk-SNARKs, to produce a proof that confirms the validity and compliance of the recorded data without disclosing the data itself. The goal is to ensure that the information, such as the location, time, and quantity of the catch, can be verified by other parties without direct access to the underlying values. This is essential to maintain confidentiality while supporting verifiable claims.

Privacy remains a primary consideration throughout this process. Sensitive details, including location references and identities of the parties involved, are protected within the structure of the proof. This privacy-focused approach is consistent with the fundamental design of ZKPs and is key to their practical use in secure and regulation-compliant seafood traceability systems.

In addition, compliance with sustainability requirements is incorporated into the proof generation process.

This includes verification that the seafood catch adheres to established industry-specific guidelines, such as sustainable fishing practices, catch quotas, and regulatory conservation standards. These checks ensure that the catch aligns with environmental and legal expectations without disclosing sensitive operational data.

Once the ZKPs is generated, it is formatted for compatibility with the SCs interface. The next essential phase involves integration with SCs deployed on a blockchain platform. These SCs are specifically designed to accept ZKPs submitted by fishermen or seafood suppliers and to execute automated verification logic. Upon submission, the SCs validates the proof to confirm its correctness and compliance, thereby ensuring trust and transparency without compromising confidentiality.

If the proof is successfully verified, the SCs records the associated catch data along with its verification status onto the blockchain ledger. This ledger, known for its immutability and transparency, plays a pivotal role in enabling end-to-end traceability and enforcing accountability throughout the seafood supply chain.

The verified information is then made accessible to authorized stakeholders, including consumers, regulatory bodies, and participants in the supply chain. Through blockchain infrastructure, these entities can trace the provenance, authenticity, and compliance of the seafood product, ensuring compliance with legal and sustainability standards during harvesting, without exposing sensitive data. This privacy-preserving feature is reinforced through the application of the Pedersen Commitment Scheme, which provides strong guarantees for data confidentiality and integrity. Collectively, this process enhances transparency, reinforces stakeholder trust, and strengthens the credibility of the seafood industry.

To ensure compliance, privacy, and authenticity at the point of seafood capture, the proposed model integrates ZKPs into the catching process, as described in Algorithm 2. This ZKPs-based mechanism allows catch data such as time, location, species, quantity, and environmental conditions to be validated without revealing sensitive details. The process includes the generation of cryptographic proofs, the verification of sustainability compliance and the verification based on smart contracts, ensuring that only legitimate and regulation-compliant records are stored on the blockchain.

Algorithm 2 ZKPs-Based Seafood Catching Process.

Require: Input set $\mathcal{C} = \{t, l, s, q, \lambda, \epsilon\}$ where: t = timestamp, l = location coordinates, s = species, q = quantity, λ = license set, ϵ = environmental parameters

Ensure: Validated catch record ρ appended to blockchain ledger $\mathcal{B}$ with zero-knowledge compliance proof π

Step 1: Formalized Data Acquisition

For each seafood catch event e_i, collect:

$$\mathcal{C}_i = \{t_i, l_i, s_i, q_i, \lambda_i, \epsilon_i\}$$

Store $\mathcal{C}_i$ securely off-chain for preprocessing and constraint validation

Step 2: ZKPs Generation

Construct constraint system $\mathcal{S}_{\text{catch}}(\mathcal{C}_i)$ such that:

$$\mathcal{S}_{\text{catch}}(\mathcal{C}_i) = 1 \iff \text{Catch is legal and accurate}$$

Generate zk-SNARKs proof π_i satisfying:

$$\text{Prover } \mathcal{P} \to \mathcal{V} : \pi_i = \text{ZKPs}(\mathcal{S}_{\text{catch}}, \mathcal{C}_i)$$

Use commitment scheme (e.g., Pedersen Commitment) to hide $\mathcal{C}_i$ while proving correctness.

Step 3: Sustainability Compliance Verification

Evaluate sustainability constraint $\mathcal{S}_{\text{sust}}(\mathcal{C}_i) = 1$

Integrate $\mathcal{S}_{\text{sust}}$ within the composite ZKPs system:

$$\pi_i \text{ encodes } \mathcal{S}_{\text{catch}} \wedge \mathcal{S}_{\text{sust}}$$

Step 4: ZKPs Packaging for SCs Submission

Encode ZKP π_i and metadata $m_i = h(\mathcal{C}_i)$ for compatibility with SCs input format

Form submission packet $\mathcal{P}_i = (\pi_i, m_i)$

Step 5: SCs-based Verification on Blockchain

Submit $\mathcal{P}_i$ to deployed SCs:

$$\text{SCs}_{\text{catch}}(\pi_i, m_i) \to \text{accept/reject}$$

SCs performs on-chain verification:

$$\mathcal{V}(\pi_i) = \begin{cases} 1 & \text{if proof is valid and compliant} \\ 0 & \text{otherwise} \end{cases}$$

Step 6: Ledger Update

if $\mathcal{V}(\pi_i) = 1$ **then**

    Append validated record $\rho_i = \{\pi_i, m_i, \text{status} = 1\}$ to blockchain $\mathcal{B}$

else

    Flag record for manual inspection, $\rho_i \to \text{status} = 0$

end if

Step 7: Stakeholder Access and Provenance Query

Grant view access to authorized users $u_j \in \mathcal{U}$:

$$\text{Query}_{u_j}(\mathcal{B}) \to \{\rho_i \mid \text{verified}\}$$

Expose only non-sensitive commitments, preserve privacy via ZKPs commitment hiding.

Step 8: Integrity and Trust Assurance

Use cryptographic commitment scheme (e.g., Pedersen) to bind m_i with integrity.

Ensure end-to-end transparency and trust by maintaining publicly verifiable logs in $\mathcal{B}$.

3.2 ZKPs-based seafood reacquisition process

In this subsection, we present the detailed design of the proposed ZKPs-based seafood reacquisition process. As illustrated in Fig. 3, the framework introduces a novel approach to addressing the challenges of secure traceability and verification of returned seafood within a complex supply chain environment.

At its foundation, the framework integrates BT with a confidential ZKPs mechanism to preserve both the integrity and privacy of the seafood reacquisition workflow. The process begins with the deployment of SCs on a blockchain platform, where a set of predefined conditions governing seafood returns is encoded. These conditions include key verification criteria such as origin authentication, product condition evaluation, and documented justification for the return.

When seafood products are returned to distributors, essential data is collected, including information about the source, physical condition, and reason for return. This information is then securely transformed into a confidential ZKPs using cryptographic methods. The ZKPs allows validation of the return event against the predefined conditions without revealing any sensitive details, thereby ensuring data privacy, a core principle of the proposed system.

The next phase in the process involves submitting the ZKPs, along with relevant information about the returned seafood, to the previously deployed SCs on the blockchain platform. At this stage, the core functionality of ZKPs is utilized. The SCs are designed to verify the submitted proofs against the predefined reacquisition conditions without revealing any sensitive data. This verification process rigorously evaluates whether the returned seafood complies with the established criteria, such as legal origin and adherence to accepted return protocols.

Depending on the outcome of the ZKPs verification and the encoded conditions, the SCs proceeds along one of two logical paths. If the proof is valid and all conditions are satisfied, the SCs automatically approves the reacquisition, confirming the legitimacy of the return. Conversely, if the verification fails or any of the conditions are not met, the SCs rejects the reacquisition request, thereby preventing non-compliant or unauthorized return transactions.

A key feature of the proposed framework lies in its robust record-keeping capability. Upon approval of a seafood reacquisition transaction, the system systematically records the event on the blockchain, generating a transparent and immutable ledger entry. This process ensures traceability and accountability throughout the seafood supply chain while supporting regulatory compliance. The SCs validate each reacquisition through ZKPs, enabling verification without revealing underlying data and thereby preserving the confidentiality of all parties involved. Distributors benefit from assurance that sensitive information related to returned seafood remains protected, while consumers and stakeholders gain the ability to independently verify product authenticity and provenance. By maintaining a secure, verifiable audit trail, the framework enhances transparency, promotes trust, and reinforces confidence across the entire supply chain ecosystem.

Algorithm 3 outlines the seafood reacquisition process, which integrates ZKPs and SCs to verify returned items without disclosing sensitive details. The framework verifies the legitimacy of each return by checking its source, condition, and justification against predefined criteria. Validated returns are securely recorded on the

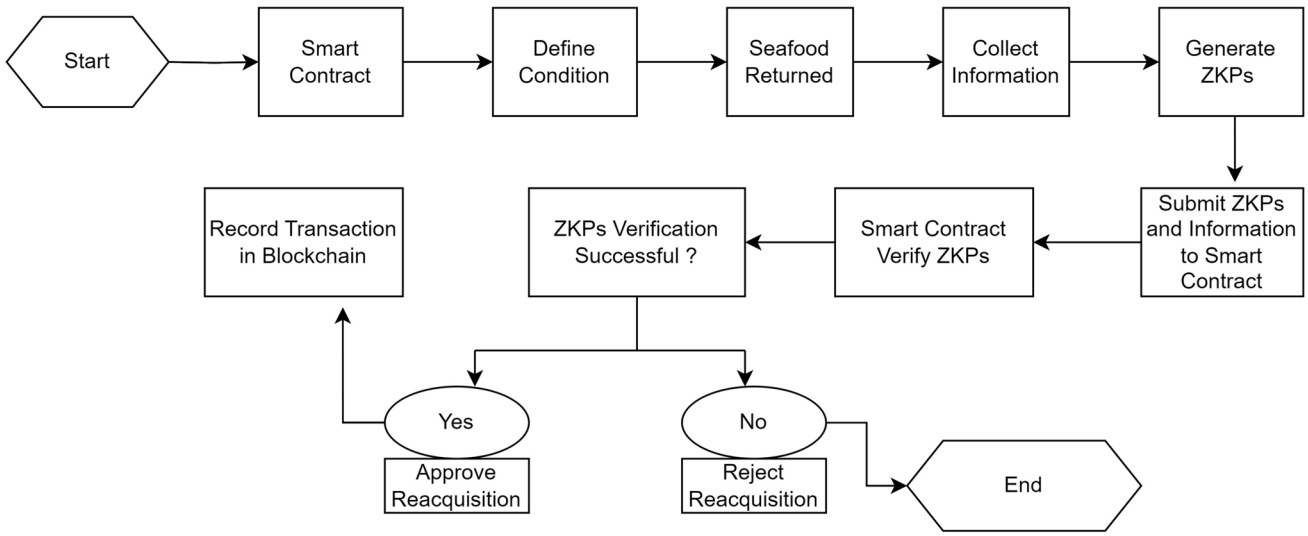


Fig. 3 ZKPs-Based Seafood Reacquisition Process

blockchain ledger, ensuring transparent tracking while maintaining data confidentiality and regulatory accountability across the supply chain. This approach helps prevent fraudulent returns and ensures that only compliant and quality-controlled seafood enters the secondary supply loop. Furthermore, the immutable audit trail supports traceability, enabling stakeholders to validate each reacquisition event with cryptographic assurance.

Algorithm 3 ZKPs-based Seafood Reacquisition Process.

Require: Reacquisition data $\mathcal{R} = \{r_s, r_c, r_r\}$ where r_s = source, r_c = condition, r_r = return reason; SCs, Blockchain ledger $\mathcal{B}$, ZKPs
Ensure: Validated reacquisition record ρ_r stored on $\mathcal{B}$
Step 1: SCs Initialization
Define SCs $\mathcal{SC}_r$ with constraint system $\mathcal{C}_r$ such that:

$$\mathcal{C}_r(r_s, r_c, r_r) = 1 \iff \text{Reacquisition conditions satisfied}$$

Constraints include:

- $r_s \in \Sigma_s$: source validity
- $r_c \in \Gamma_c$: condition within threshold θ
- $r_r \in \Lambda_r$: return justification

Step 2: Data Collection and ZKPs Construction
Collect tuple $\mathcal{R}_i = \{r_s^i, r_c^i, r_r^i\}$ for return event i
Construct ZKPs π_r^i:

$$\pi_r^i \leftarrow \text{Prove}\,(\mathcal{C}_r(\mathcal{R}_i) = 1)$$

Use commitment scheme $\text{Com}(\mathcal{R}_i)$ to preserve data privacy
Step 3: SCs Interaction
Submit $(\pi_r^i, h(\mathcal{R}_i))$ to $\mathcal{SC}_r$
SCs performs on-chain verification:

$$\mathcal{V}(\pi_r^i) = \begin{cases} 1 & \text{if } \pi_r^i \text{ satisfies } \mathcal{C}_r \\ 0 & \text{otherwise} \end{cases}$$

Step 4: Verification Outcome
if $\mathcal{V}(\pi_r^i) = 1$ **then**
  Approve reacquisition, set $\rho_r^i = \{\pi_r^i, h(\mathcal{R}_i), \text{status} = 1\}$
  Append ρ_r^i to blockchain: $\mathcal{B} \leftarrow \mathcal{B} \cup \rho_r^i$
else
  Reject reacquisition, log ρ_r^i with status $= 0$
end if
Step 5: Ledger Integrity and Auditability
Maintain $\mathcal{B}$ as an immutable audit trail:

$$\forall \rho_r^i \in \mathcal{B},\ \text{integrity}(\rho_r^i) = \text{true}$$

Step 6: Privacy Preservation
All evaluations rely on π_r^i, preserving confidentiality of $\mathcal{R}_i$
No plaintext data exposed to $\mathcal{SC}_r$ or third parties
Step 7: Stakeholder Transparency
Authorized parties $\mathcal{U}$ access proofs:

$$\forall u_j \in \mathcal{U}, \text{Query}_{u_j}(\mathcal{B}) \rightarrow \rho_r^i$$

Provide seafood provenance verification without revealing private inputs
Step 8: Regulatory Compliance
Use $\mathcal{B}$ to demonstrate compliance:

$$\exists \rho_r^i \Rightarrow \text{compliance}(\rho_r^i) = \text{valid}$$

Ensure accountability for all return events

3.3 Security against cyberattacks and tampering

Blockchain strengthens the security of the proposed seafood requisition framework through cryptographic immutability and distributed consensus. Each block B_i contains a hash $H(B_i)$, computed as in Eq. 1.

$$H(B_i) = \text{Hash}(B_{i-1} \parallel D_i \parallel T_i) \tag{1}$$

where B_{i-1} is the hash of the previous block, D_i denotes the transaction data, and T_i is the timestamp. This hash-chaining mechanism ensures that any alteration in prior blocks propagates inconsistencies forward, thereby invalidating the entire chain. Such tampering is computationally infeasible due to the effort required to recompute all subsequent hashes and gain majority consensus.

To append a new block B_{n+1}, a consensus function $\mathcal{C}$ must be satisfied across a network of N nodes, see Eq. 2.

$$\mathcal{C}(B_{n+1}) = \text{True} \quad \text{if} \quad \sum_{j=1}^{N} \delta_j \geq \left\lceil \frac{N}{2} + 1 \right\rceil \tag{2}$$

where $\delta_j = 1$ if node j approves the block, and 0 otherwise. This majority rule prevents unauthorized modifications unless an adversary controls more than 50% of the network nodes (i.e., a 51% attack), which is computationally and economically impractical in a properly decentralized system.

SCs $\mathcal{SC}_k : \mathcal{X} \rightarrow \mathcal{Y}$ embedded on-chain execute deterministic logic as described in Eq. 3.

$$\mathcal{SC}_k(x) = y \quad \text{only if} \quad x \in \text{ValidDomain} \tag{3}$$

ensuring that operations are automatically enforced without manual intervention. This architectural design minimizes the attack surface, enhances data authenticity, and ensures robust protection against cyber threats such as data tampering, denial-of-service, and unauthorized access.

4 Implementation and result analysis

The technical implementation utilizes remix IDE, metamask, and the sepolia testnet due to their ease of integration, accessibility, and suitability for rapid prototyping. Remix IDE, chosen over alternatives like the truffle suite, offers a browser-based environment with a user-friendly interface, real-time compilation, and built-in deployment features—ideal for efficient SCs development and testing. Metamask, serving as the wallet interface, ensures secure key management and smooth interaction with ethereum-based networks.

It allows users to easily switch between multiple networks, including testnets and mainnets, which is essential for verifying contract behavior under different conditions. Additionally, its browser extension provides a seamless connection between the frontend and blockchain, enabling real-time transaction execution and monitoring. The sepolia testnet, preferred over goerli, provides faster block times, lower network congestion, and enhanced reliability, aligning well with the performance and testing requirements of the proposed framework. These tools also support solidity versioning and debugging capabilities, streamlining the contract validation process. Their wide adoption and community support contribute to faster troubleshooting and learning. In combination, this toolset enables secure, and cost-effective deployment of decentralized applications during the development phase.

Moving forward, we discuss the outcomes based on our proposed framework findings. This result will provide a comprehensive insight into the implications and importance of our proposed findings. Table 1 presents comprehensive details about the Ethereum blockchain transaction associated with the deployment of the Requisition SCs. The information includes elements such as the transaction hash, which uniquely identifies the transaction on the blockchain, and block specifics, including the block number and timestamp indicating when the transaction was confirmed. It also details gas-related parameters such as the gas price (the cost per unit of computational effort), gas limit (the maximum computational units allowed for the transaction), and gas used (the actual amount consumed), along with the transaction fee, calculated as the product of gas used and gas price. Additionally, the table includes the transaction type, which denotes whether the transaction follows the legacy or EIP-1559 fee structure, the nonce, representing the transaction count from the sender's address to ensure proper sequencing, and the block position, which indicates the transaction's order within the block. Together, these elements offer a detailed and verifiable overview of the requisition SCs deployment on the ethereum blockchain.

The information in Table 2 primarily focuses on consensus and blockchain infrastructure aspects. The requisition SCs are deployed and executed within specific blocks on the ethereum blockchain, each identified by a unique block number and timestamp. Consensus-related details, such as the slot (a fixed time interval used to propose a new block) and epoch (a group of consecutive slots), help ensure the secure and orderly execution of transactions across the network. The slot root hash, a cryptographic fingerprint of the blockchain state at a given slot, and block randomness, which introduces unpredictability into the block selection process, both contribute to the integrity and fairness of the consensus mechanism. Additionally, the Beacon Chain Deposit Count reflects the number of validator deposits made into the ethereum 2.0 network, serving as an indicator of validator participation and overall network health. These parameters collectively provide insight into the operational status, security, and decentralization level of the ethereum infrastructure supporting the requisition SCs.

The contents of Table 3 are specifically tailored to shed light on the intricacies surrounding the deployment and execution of the Requisition SCs within the Ethereum ecosystem. The Block Hash and Parent Hash serve as unique identifiers, ensuring the cryptographic integrity of the block where the Requisition SCs transactions occur. StateRoot and WithdrawalsRoot reflect the final state of the Ethereum world after transactions, including potential changes caused by the Requisition SCs. The Block Reward incentivizes miners to include Requisition-related transactions, contributing to the blockchain's security. Total Difficulty showcases the cumulative computational effort within the blockchain, enhancing the overall resilience of the Requisition SCs. The

Table 1 Reacquisition SCs Deployment Result Analysis

Parameter	Result
Transaction Hash	0x161fcec21e5fdb79375f893ad-5b0767869319733580f220d7b 91c0afe84de192
Block NO.	5161969
Block Hash	0x4ff5cb92c0bc5201d-717bbd42162f78ac2c9cac88f52774 ce406a609e14b508b
Transaction Fee	0.003011657402979866 ETH
Gas Price	1.600724449 Gwei (0.000000001600724449 ETH)
Gas Limit and Usage by Txn	1,897,533 and 1,881,434 (99.15%)
Gas Fees	Base: 0.100724449 Gwei, Max: 1.669390535 Gwei, and Max Priority: 1.5 Gwei
Txn Type	2 (EIP-1559)
Nonce	77
Block Position	10

Table 2 Focuses on Consensus and Blockchain Infrastructure, Particularly Examining the Requisition SCs

Parameter	Result
Slot	4216542
Epoch	131766
Proposer Index	587
Slot Root Hash	0xe5712e6aefe55cca55717172ea-42bec8eedc365c1e43 e4b533ec4c3dfa31d4b4
Beacon Chain Deposit Count	403
Slot Graffiti	Besu Teku W3S (Hex:0x4265737520f09f909020546 56b7 52057335300000000000000000000000000000)
Block Randomness	0xc5a2d6f742e801bcd5222a2bb7e2976c9c59e9df9 2edfa2504d8c5a01308d144

Table 3 Elucidates Intricacies of Deploying and Executing the Requisition SCs in Ethereum

Parameter	Result
Hash	0x4ff5cb92c0bc5201d-717bbd42162f78ac2c9cac88f52 774ce406a609e14b508b
Parent Hash	0x1e97167ff93e802f54bb8235ab9e-6f4a3bbe451858098 6ccfe9b9c49676af881
StateRoot	0x98fed32109e361f3b861beca77 832a13fb-c8e6e392e6a245c0866d cb16915d90
WithdrawalsRoot	0xf298dc29e553d61b8e66f2348077 c8649a0bb8bab69ef2b7093841075065e5fb
Block Reward	0.008561613059343462 ETH
Total Difficulty	17,000,018,015,853,232
Size	48,651 bytes
Gas Used	7,771,129(25.90%)
Base Fee Per Gas	0.000000000100724449 ETH (0.100724449 Gwei)
Burnt Fees	0.000782742686632921 ETH

block's Size indicates the space occupied by transactions, including those involving the deployment and execution of the Requisition SCs. Gas Used and Base Fee Per Gas directly influence transaction costs, encompassing activities related to the Requisition SCs. Finally, Burnt Fees contribute to Ethereum's deflationary mechanism, impacting the economic context within which the Requisition SCs operate.

The Table in 4 presents a comprehensive analysis of gas costs incurred within the ethereum environment, highlighting key components associated with the deployment and execution of SCs. Specifically, it outlines the code deposit cost, execution cost, and the total cumulative cost, offering valuable insights into the computational resources and financial implications of SCs operations. The code deposit cost of 1,673,600 denotes the gas consumed during the contract's deployment phase, which includes storing the contract's bytecode on the blockchain—a one-time cost necessary for making the contract accessible to the network. The execution cost of 26,020 represents the gas used to trigger and perform the contract's logic during a function call, reflecting the dynamic nature of resource consumption based on user interactions or internal logic. Combined, the total cost of 1,699,620 encapsulates the entire computational effort and corresponding financial expense required for both deployment and execution.

This detailed breakdown is useful for developers, auditors, and researchers aiming to optimize SCs design by identifying cost-intensive operations. Furthermore, understanding gas utilization supports more accurate budgeting and resource planning in environments where computational cost is a concern. By analyzing such granular gas metrics, stakeholders can make informed decisions regarding contract efficiency and upgrade strategies within ethereum-based systems.

Table 4 Gas Costs Associated with Code Deposit, Execution, and the Overall Total Cost

Item	Gas Cost
codeDepositCost	1673600
executionCost	26020
totalCost	1699620

Table 5 presents a comprehensive analysis of a transaction involving the deployment of the seafood ZKPs of Catching SCs. The recorded data includes the transaction hash, associated block number, transaction fee, transaction type (EIP-1559), nonce, block position, block hash, gas limit, gas used, and detailed gas fee components, including base fee and priority fee, along with the effective gas price. This information is essential for understanding the computational cost and structural characteristics of smart contract interactions on the ethereum blockchain. Moreover, analyzing these parameters helps in evaluating network behavior under different fee models and optimizing gas usage in similar SCs deployments. Such insights support informed decision-making in terms of contract design, fee estimation, and network interaction efficiency.

The parameters in Table 6 for Block number 5162741 offer valuable insights into the operation of the seafood ZKPs of Catching SCs on the Ethereum blockchain. These include cryptographic identifiers (such as block hash and parent hash), the contract's current execution state, withdrawal activity, reward distribution, computational difficulty, block size, gas limits and usage, and associated economic indicators. Collectively, these metrics provide a snapshot of SCs behavior and resource consumption within the specified block. This detailed view supports analysis of the contract's performance in terms of security, transparency, and operational efficiency within the blockchain environment.

Table 7 highlights consensus-related parameters associated with the seafood ZKPs of Catching SCs on the ethereum blockchain. In Slot 4217368 of Epoch 131792 (Block

Table 5 Analyzes a Transaction with a Seafood ZKPs of Catching SCs

Parameter	Result
Transaction Hash	0x9a53b5ae0c7d0d76c81c3f7 ae88aa-c7ab93829c7616dfaf2 9337a7adcaee2234
Block Number	5162741
Transaction Fee	0.00379614236964984 ETH
Txn Type	2 (EIP-1559)
Nonce	81
Block Position	65
Block Hash	0x475b0b8921d5429a53d4b2274dac8495 50da8c0a2e3506f81aa4f74814e3adb7
Gas Limit and Usage by Txn	2,337,675 and 2,318,110 (99.16%)
Gas Fees	Base: 0.137602344 Gwei, Max: 1.653075757 Gwei, and Max Priority: 1.5 Gwei
Gas Price	1.637602344 Gwei (0.000000001637602344 ETH)

Table 6 Parameters for Block 5162741 Offer Key Insights into the Seafood ZKPs Catching SCs on Ethereum

Parameter	Result
Hash	0x475b0b8921d5429a53d4b2274dac849550da8c 0a2e3506f81aa4f74814e3adb7
Parent Hash	0x6a287ee5b2f9e5a71aa27e5 7db62361d7184b5b77b26cc2445cfb01078013a7a
StateRoot	0x61fa84f05cd3e27de22d3839 191eb2ff11c52c2a87ca728ffa07652ce76cb32f
WithdrawalsRoot	0xdfe9ebb3d87d2c6c5f3ed299 7b25dd8d-b800a8f5c 5dfa91886f1a590143b6a8d
Block Reward	0.043501714751107842 ETH
Total Difficulty	17,000,018,015,853,232
Size	136,857 bytes
Gas Used	24,884,345(82.95%)
Base Fee Per Gas	0.000000000137602344 ETH (0.137602344 Gwei)
Burnt Fees	0.00342414420090468 ETH

Table 7 Highlights the Consensus Mechanisms and Interactions of the Seafood ZKPs Catching SCs

Parameter	Result
Slot	4217368
Epoch	131792
Proposer Index	1934
Slot Root Hash	0xbbe80d6b7729a84c-8765d25a94e4a75e039 e9ef79951c80902ba29c9d8ffc069
Beacon Chain Deposit Count	403
Slot Graffiti	0x (Hex:Null)
Block Randomness	0x6860a757ac9847952d0b353c1909823f 46a6e6f2c0e917dbd4e778823662b0ec

5162741), the Proposer Index (1934) identifies the block proposer, while the Slot Root Hash ensures data integrity. The Beacon Chain Deposit Count (403) reflects validator participation in ethereum 2.0, and the Block Randomness value supports cryptographic unpredictability. These parameters collectively represent the secure and coordinated network environment in which the SCs operates.

4.1 Snapshots of the seafood reacquisition process in the proposed framework

The provided data consists of Ethereum event logs related to a system associated with seafood quality verification and reacquisition. Each entry in the array represents a distinct Ethereum event log, and these logs are associated with specific events and proofs within the system. In the first event log in Fig. 4, a "ConfidentialQualityVerificationProof" event is recorded. It originates from the Ethereum address '0x6ecf-97315c15acc2e136ecf046c90ee6169fc98b'. The event is identified by address '0x5a10fe8d9cda5c276d976da25b-d5e8152c6b9f6012f17d71c5c075d4a6a81648'. The associated arguments include a serial number (12) and a quality proof (0xad8aec7171b7cf5b1c4370a479fbe496747764f8d-76b844bc703afd0ef9d1986).

Following that, the second event log in Fig. 5 captures an "AnonymousReacquisitionInitiatorProof" event. Again, originating from the Ethereum address '0x6ecf-97315c15acc2e136ecf046c90ee6169fc98b', this event is identified by address '0x036ba29b11bb3cadf8d287f2d3a-1f8a9745d143a6d1e8fcc95860e786faba5ea'. The arguments contain a serial number (12) and an initiator proof (0xede8db856c9460ab9aaf01a00714fe6d5769734acb7a1b-dcd23d9f217ff2fb9d).

The third event log in Fig. 6 corresponds to a "Secure-PackagingIntegrityProof" event, once again originating from the Ethereum address '0x6ecf97315c15acc2e136ecf-046c90ee6169fc98b'. The topic for this event is '0x7ed-31447c025fb4285f0ec801217af34dd9433123c6a6a3b-



Fig. 4 Log of the Proposed Framework: ConfidentialQualityVerificationProof



Fig. 5 Log of the Proposed Framework:AnonymousReacquisitionInitiatorProof



Fig. 6 Log of the Proposed Framework:SecurePackagingIntegrityProof

660cb2570b784b22'. The associated arguments include a serial number (12) and a packaging integrity proof (0x341aa32a29fbc17b9fb2886c0e702b21d18f6b9e58e-87ca80eb55a60fa614205).

In the fourth event log in Fig. 7, a "SelectiveInformationDisclosureProof" event is recorded, originating from the same Ethereum address. The topic for this event is '0x4b17040e84d4fda5c61301885458c4d32673d-f0616315a57f25d75727a8391c7'. The arguments include a serial number (12) and a disclosure proof (0xed4c2566576b-853c4090bfc41ae477e2f1f46cdc7c63d9296004a12ce2a-b1eb7).

```
        {
                "from": "0x6ecf97315c15acc2e136ecf046c90ee6169fc98b",
                "topic": "0x4b17040e84d4fda5c61301885458c4d32673df0616315a57f25d75727a8391c7",
                "event": "SelectiveInformationDisclosureProof",
                "args": {
                        "0": "12",
                        "1":
"0xed4c2566576b853c4090bfc41ae477e2f1f46cdc7c63d9296004a12ce2ab1eb7",
                        "serialNumber": "12",
                        "disclosureProof":
"0xed4c2566576b853c4090bfc41ae477e2f1f46cdc7c63d9296004a12ce2ab1eb7"
                }
        },
        {
                "from": "0x6ecf97315c15acc2e136ecf046c90ee6169fc98b",
                "topic": "0xb46e48bede26902470aeb1512c9fb77c7d4aa21afbdd0b6f17b191847de95c94",
                "event": "SustainableReacquisitionProof",
                "args": {
                        "0": "12",
                        "1":
"0x575216413f7c502eed144bfdbe23d40e09aab84dbf92c2d77f224177f6a43d5d",
                        "serialNumber": "12",
                        "sustainabilityProof":
"0x575216413f7c502eed144bfdbe23d40e09aab84dbf92c2d77f224177f6a43d5d"
                }
        },
```


Fig. 7 Log of the Proposed Framework: SelectiveInformationDisclosureProof

The fifth event log in Fig. 8 captures a "SustainableReacquisitionProof" event from the Ethereum address '0x6ecf97315c15acc2e136ecf046c90ee6169fc98b'. The associated topic is '0xb46e48bede26902470aeb1512c9fb77c7d4aa21afbdd0b6f17b191847de95c94'. The arguments include a serial number (12) and a sustainability proof (0x575216413f7c502eed144bfdbe23d40e09aab84dbf92c2d77f224177f6a43d5d).

Lastly, the sixth event log in Fig. 9 signifies a "ReacquisitionInitiated" event. Originating from the Ethereum address '0x6ecf97315c15acc2e136ecf046c90ee6169fc98b', this event is identified by the topic '0xee5c3bc7c4144d5799144c52499cf85cc03994642d4d0cff0a083ff4c5ac5d44'. The arguments contain a serial number (12) and information about the initiator (0x261846E6E3ABed5d5C57eD1B63dA1cE3f78a2eef).

In summary, these Ethereum event logs document various aspects of a seafood quality verification and reacquisition system, including proofs related to confidentiality, initiator details, packaging integrity, information disclosure,


```
        },
        {
                "from": "0x6ecf97315c15acc2e136ecf046c90ee6169fc98b",
                "topic": "0xee5c3bc7c4144d5799144c52499cf85cc03994642d4d0cff0a083ff4c5ac5d44",
                "event": "ReacquisitionInitiated",
                "args": {
                        "0": "12",
                        "1": "0x261846E6E3ABed5d5C57eD1B63dA1cE3f78a2eef",
                        "serialNumber": "12",
                        "initiator": "0x261846E6E3ABed5d5C57eD1B63dA1cE3f78a2eef"
                }
        }
]
```


Fig. 8 Log of the Proposed Framework: SustainableReacquisitionProof

```
[
        {
                "from": "0x6ecf97315c15acc2e136ecf046c90ee6169fc98b",
                "topic": "0x5a10fe8d9cda5c276d976da25bd5e8152c6b9f6012f17d71c5c075d4a6a81648",
                "event": "ConfidentialQualityVerificationProof",
                "args": {
                        "0": "12",
                        "1":
"0xad8aec7171b7cf5b1c4370a479fbe496747764f8d76b844bc703afd0ef9d1986",
                        "serialNumber": "12",
                        "qualityProof":
"0xad8aec7171b7cf5b1c4370a479fbe496747764f8d76b844bc703afd0ef9d1986"
                }
        },
        {
                "from": "0x6ecf97315c15acc2e136ecf046c90ee6169fc98b",
                "topic": "0x036ba29b11bb3cadf8d287f2d3a1f8a9745d143a6d1e8fcc95860e786faba5ea",
                "event": "AnonymousReacquisitionInitiatorProof",
                "args": {
                        "0": "12",
                        "1":
"0xede8db856c9460ab9aaf01a00714fe6d5769734acb7a1bdcd23d9f217ff2fb9d",
                        "serialNumber": "12",
                        "initiatorProof":
"0xede8db856c9460ab9aaf01a00714fe6d5769734acb7a1bdcd23d9f217ff2fb9d"
                }
```


Fig. 9 Log of the Proposed Framework: ReacquisitionInitiated

and sustainability. Each log provides specific information about the events and proofs associated with the system's processes.

4.2 Simulation analysis

In our simulation experiments, the system underwent varying loads ranging from 100 to 1900 concurrent users, representing seafood distributors and harvesters. Each virtual user continuously sent transactions for a fixed duration, emulating seafood requisition requests within the blockchain-based supply chain network. The test environment maintained a stable network connection with an average bandwidth of 100 Mbps. The SCs deployment took place on the sepolia testnet, with JMeter configured to communicate through a custom HTTP endpoint. This setup enabled the measurement of key performance metrics under increasing load conditions. This setup enabled consistent measurement of performance metrics such as throughput, latency, response time and STD. deviation. This configuration reflects real-world usage scenarios and enables reproducibility and validation by other researchers.

Throughput, which measures the total number of distributor and harvester interactions within a specific time window, emerged as a critical parameter of interest. The objective, to analyze how the blockchain network performs under varying seafood reacquisition and harvesting transaction loads generated by different user sets.

Latency, the time interval between initiating seafood reacquisition and harvesting requests and their inclusion in the ethereum blockchain network underwent careful evaluation.

Moreover, We examined response time, defined as the duration between the initiation of a request or action and the receipt or completion of a corresponding response. This metric provided valuable insights into the performance of the proposed model in processing transactions, highlighting its effectiveness under varying loads from distributors and harvesters.

Additionally, STD. deviation analysis is used to assess the consistency and stability of response times across multiple requests or transactions. It provides valuable insights into the performance reliability of the proposed model. A low standard deviation indicates that response times are closely clustered around the mean, reflecting consistent and predictable system behavior. In contrast, a higher standard deviation suggests greater variability in response times, which may indicate performance fluctuations or irregularities. This analysis supports the optimization of the model to ensure stable and reliable execution under varying operational conditions.

In, Figs. 10, and 11, we present performance data related to throughput, latency, response time, and STD. deviation for the proposed seafood reacquisition SCs for distributors and the proof of catching SCs for harvesters. Notably, as the number of distributors and harvesters increases, the throughput initially demonstrates a steady rise. However, beyond a certain threshold, throughput begins to fluctuate, indicating potential scalability limits or network congestion effects. In contrast, latency, response time and STD. deviation maintain a consistent low level with only minor variations throughout the entire transaction.

Figure 12 provides a visual representation of essential metrics. The x-axis denotes the number of distributors and harvesters, while the y-axis showcases throughput (measured in TPS), latency, response time (measured in milliseconds), and STD. Deviation. A noteworthy observation is

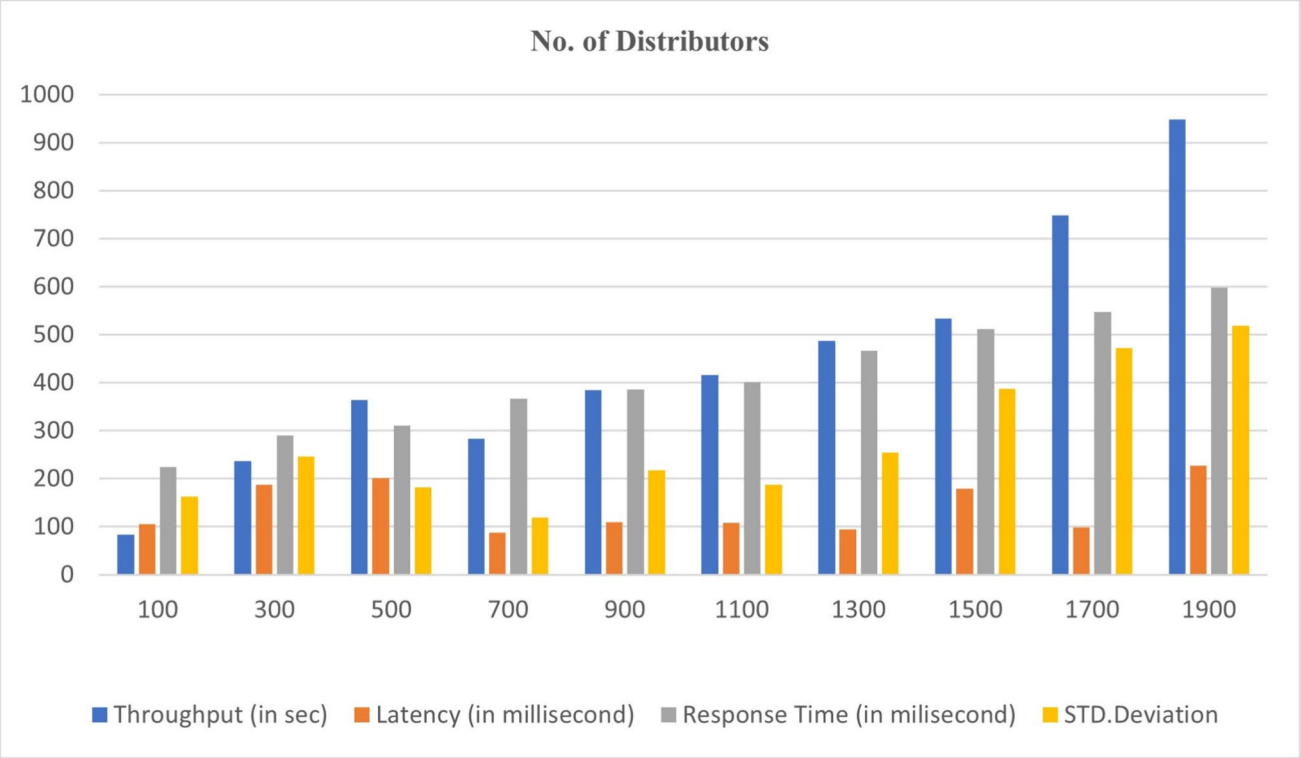


Fig. 10 Assessment of the Reacquisition SCs Performance Metrics, including Throughput, Latency, Response Time, and STD. Deviation

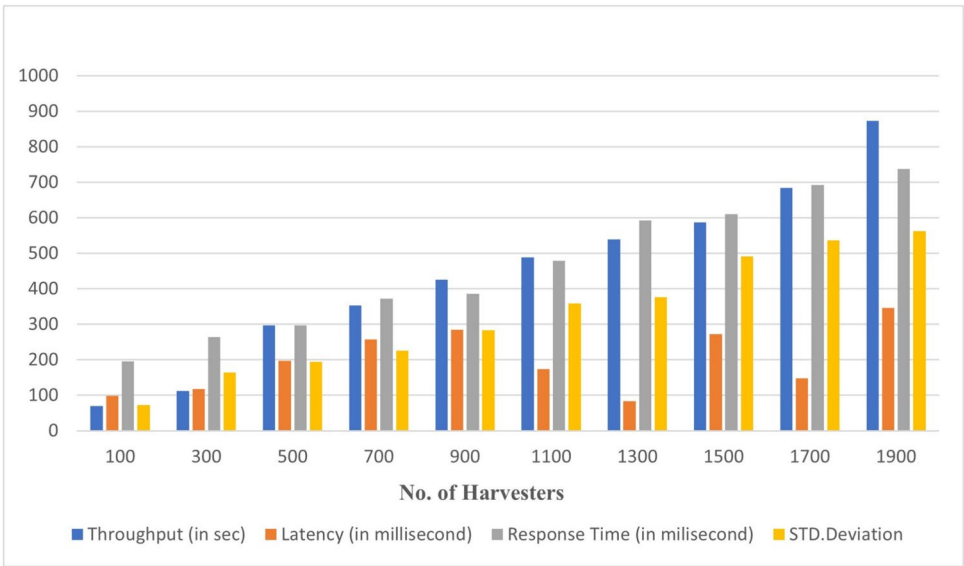


Fig. 11 Assessment of the Harvester's SCs Performance Metrics, including Throughput, Latency, Response Time, and STD. Deviation

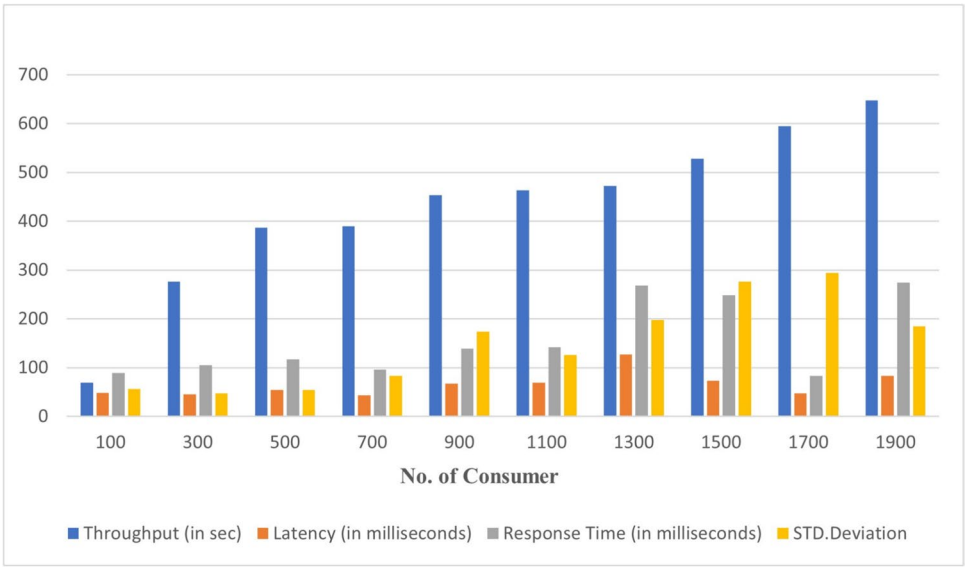


Fig. 12 Analysis of the Proposed Framework's Performance Metrics, including Throughput, Latency, Response Time, and STD. Deviation

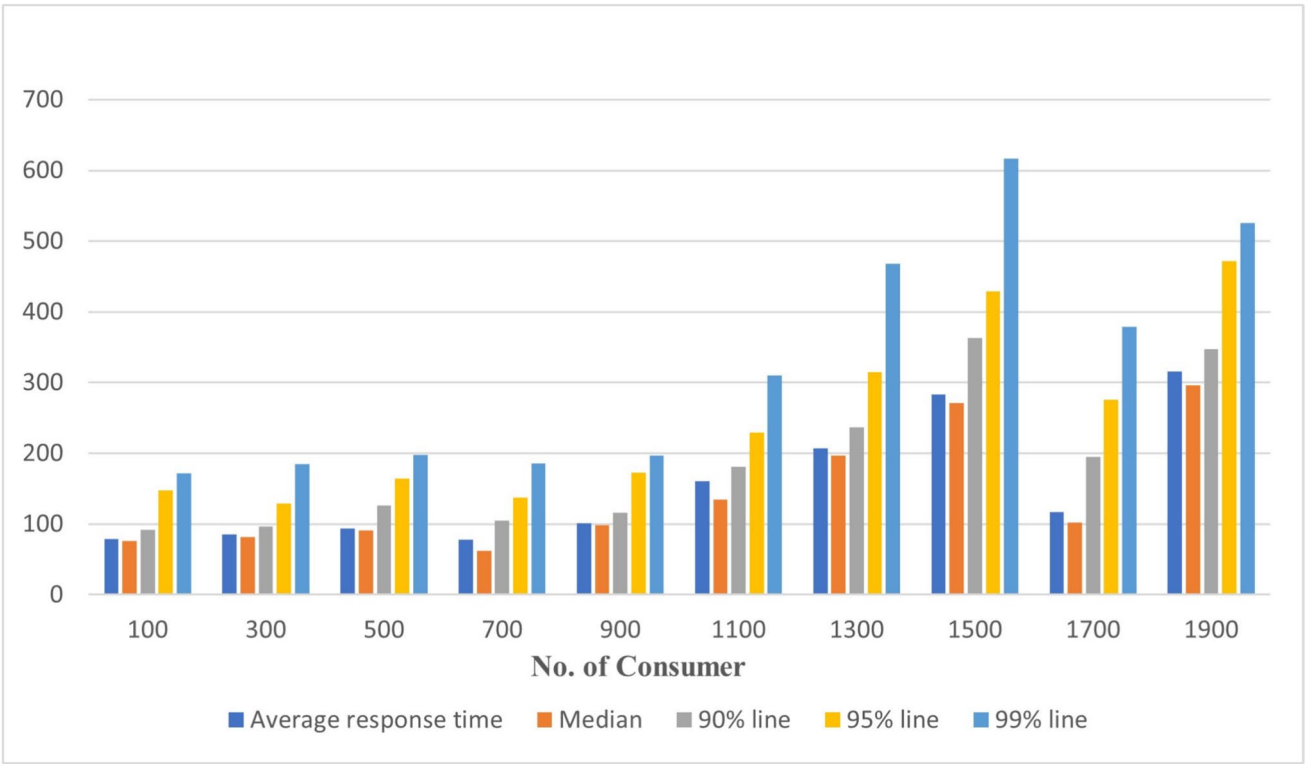


Fig. 13 Assessment of Response Time Variations in the Proposed Framework

the correlation between distributors and harvesters, count, and throughput, with an increase followed by a decrease. This underscores the model's adeptness in managing high distributor's and harvester's loads. A reduction in latency signifies enhanced transaction processing speed and system responsiveness. However, it's noteworthy that response times exhibit fluctuations, with certain transactions experiencing increases and others showcasing decreases. Based on Fig. 12, we can calculate the average values for each metric across different configurations: average throughput is 682.36 TPS, average latency is 64.1 milliseconds, and average response time is 75.5 milliseconds. Additionally, the standard deviation of the response time, used to assess the predictability of the proposed framework, is calculated as 68.7 milliseconds.

Figure 13 illustrates a test scenario involving 100, 300, 500, to 1900 concurrent seafood distributors and harvesters, providing insights into the proposed model

performance. The depicted metrics include average response time, along with the 90th, 95th, and 99th percentile response times. These metrics play a pivotal role in evaluating the system's behaviour under varying loads and pinpointing potential performance bottlenecks. The average response time offers a glimpse into the typical experience, while percentiles such as the 90th, 95th, and 99th shed light on different performance levels. The 90th percentile represents the experience of most distributors and harvesters, the 95th percentile highlights issues affecting a significant portion, and the 99th percentile identifies extreme cases and critical performance bottlenecks within the proposed framework.

5 Discussion

This section analyzes how the proposed framework addresses key limitations observed in existing state-of-the-art systems, while offering enhanced privacy, traceability, and return validation capabilities. It also presents the core findings from experimental evaluation and discusses practical challenges related to real-world deployment.

A comparative evaluation of existing literature highlights key differences in blockchain adoption, privacy integration, requisition workflows, and sustainability focus across various supply chain systems. Several studies, such as those by Adedeji et al. [45], Nadhiroh et al. [46], and Singh et al. [47], focus on requisition systems but do not leverage blockchain technology. These solutions primarily address digitalization and automation using conventional web-based architectures without providing immutable traceability or privacy protection.

In contrast, works like Sahai et al. [32] and Malik et al. [44] utilize permissioned blockchain platforms (e.g., Hyperledger Fabric, Indy) to enable traceability and privacy using zero-knowledge proofs (ZKPs) and attribute-based encryption. However, they do not address the validation of returned goods or reacquisition processes, which are critical for industries such as seafood that deal with quality-sensitive and perishable products. Similarly, Bai et al. [40], Sedlmeir et al. [41], and Abadi et al. [42] offer theoretical advancements in blockchain privacy mechanisms, including succinct ZKPs and UC-secure protocols, but these frameworks lack practical requisition or supply chain integration.

Studies like Xiong et al. [43] and Wang et al. [50] propose blockchain-based procurement and payment systems but are confined to internal enterprise workflows and do not address public traceability or sustainability. Furthermore, none of the existing blockchain-based systems provide support for return or reacquisition validation in a public, privacy-preserving manner.

In contrast to existing approaches, the proposed BT-CSRS framework presents a comprehensive and technically robust solution that addresses critical gaps in current requisition systems. Built on the ethereum public blockchain, BT-CSRS leverages the advantages of decentralization and transparency without relying on permissioned or conceptual frameworks. A key distinguishing feature of BT-CSRS is its automated requisition handling mechanism, which is executed through SCs. Unlike many existing works that either overlook requisition processes entirely or implement them manually through centralized web applications, BT-CSRS offers an on-chain, fully automated workflow that governs both the initiation and validation of requisition and return events. This reduces manual errors, mitigates delays, and ensures that all business rules are enforced with cryptographic certainty. The framework also supports conditional logic for seafood return approval, enabling dynamic handling based on product type, origin, or supplier history—features rarely present in prior models.

To ensure confidentiality of sensitive data—such as supplier identities and return justifications BT-CSRS integrates ZKPs, enabling verifiable privacy without revealing the underlying details on-chain. Unlike previous studies that overlook reverse logistics or reacquisition flows, BT-CSRS provides explicit support for confidential product returns, making it highly relevant for industries where product authenticity, integrity, and return legitimacy are paramount. Furthermore, BT-CSRS guarantees full traceability of seafood items, enabling provenance tracking from initial sourcing to reacquisition, thereby ensuring data integrity across the entire supply chain. This level of traceability extends to both forward and reverse logistics, a capability rarely supported in prior systems. The technical performance of the framework has been validated through Jmeter based testing on the ethereum testnet, demonstrating its practical feasibility in terms of transaction throughput, response time, and latency. Taken together, these characteristics establish BT-CSRS as a state-of-the-art framework that uniquely blends transparency, automation, privacy, and performance, while promoting responsible sourcing practices in alignment with sustainable seafood industry goals.

Table 8 summarizes a comparative analysis of state-of-the-art related works, against which the proposed framework is evaluated based on blockchain usage, performance analysis, privacy mechanisms, requisition support, return validation, traceability coverage, and SCs integration.

5.1 Real-World deployment challenges

While the proposed BT-CSRS model demonstrates efficient performance in a simulated environment, several real-world

Table 8 Comparative Analysis of Related Works and the Proposed Framework

Ref.	Blockchain Platform	Network Type	Requisition Handling	Privacy Mechanism	Traceability	SCs Usage	Performance Analysis
[32]	Hyperledger	Permissioned	Not addressed	ZKPs	Contamination tracking	Yes	Yes
[40]	Conceptual	Hybrid	Not addressed	ZKPs	Theoretical discussion	Conceptual	No
[41]	Conceptual	Not specified	Not addressed	Succinct ZKPs	Partial (Energy focus)	No	No
[42]	Global Ledger	Theoretical	Not addressed	UC-secure ZKPs + Timestamps	Not traceability-specific	No	No
[44]	Hyperledger	Permissioned	Identity-based Access Control	ZKPs	Yes	Yes	Yes
[43]	Custom blockchain	Private	Payment-based interaction	Secret Sharing	Partial	Yes	Yes
[45]	None (Web-based)	N/A	Manual to automated	None	No	No	Yes
[46]	None (Web-based)	N/A	Automated (RAD model)	None	No	No	Yes
[47]	None	N/A	Process optimization only	None	No	No	No
[50]	Hyperledger	Permissioned	Internal Procurement	Basic data security	Yes	Yes	Yes
[51]	Custom	Consortium	Data Acquisition Focused	Encryption	No	No	Yes
[52]	Not specified	Public Conceptual	Government requisition	Not specified	No	Yes	No
This Work	**Ethereum**	**Public**	**Automated via SCs**	**ZKPs**	**End-to-End**	**Yes – For validation & traceability**	**Yes**

challenges may impact its scalability and feasibility. These challenges can be formally expressed and analyzed using quantitative representations to provide a more rigorous evaluation.

1. On-Chain Storage Cost The cost of storing seafood requisition transactions on-chain increases proportionally with the transaction volume. Let:

- T denote the total number of transactions,
- S represent the average size of each transaction (in bytes),
- G be the gas cost per byte on the sepolia testnet.

The total on-chain storage cost can be modeled as described in Eq. 4.

$$C_{\text{storage}} = T \cdot S \cdot G \tag{4}$$

As $T \to \infty$, especially in large-scale seafood supply chains, the cost becomes prohibitive. This necessitates hybrid solutions such as off-chain storage (e.g., IPFS), with only hash references anchored on-chain.

2. Network Scalability Bottlenecks Scalability under increasing load can be modeled by the transaction processing capacity as a function of the number of concurrent users u, see Eq. 5.

$$P(u) = \frac{T_{\max}}{1 + \alpha u} \tag{5}$$

Where:

- $T_{\max}$ is the maximum theoretical throughput (in TPS),
- α is the system's sensitivity factor to concurrent load.

As u increases, throughput $P(u)$ decreases due to increased network congestion and consensus delays, potentially creating performance bottlenecks.

3. Privacy-Preserving Requisition with ZKPs To protect sensitive requisition data, the model incorporates ZKPs. Let each requisition request be defined as shown in Eq. 6.

$$\text{Req}_i = \{d_i, q_i, s_i\} \tag{6}$$

Where:

- d_i = product description,
- q_i = quantity requested,
- s_i = source or distributor ID.

A ZKPs is generated such that:

$$\mathcal{Z}(\text{Req}_i) \to \text{true} \quad \text{iff} \quad \text{Req}_i \text{ satisfies the system's policy constraints } \mathcal{P}$$

This allows validation of each request without revealing d_i, q_i, or s_i. The time complexity for processing a transaction with ZKPs is given by Eq. 7.

$$t_{\text{txn}} = t_{\text{commit}} + t_{\text{proof}} + t_{\text{verify}} \tag{7}$$

Where:

- t_{commit} = time to commit on-chain data,
- t_{proof} = time to generate the ZKPs,
- t_{verify} = time to verify the proof.

Optimizing t_{proof} and t_{verify} is essential to maintain acceptable performance during large-scale operations.

4. Regulatory Constraints Let $\mathcal{R} = \{R_1, R_2, \ldots, R_n\}$ represent regulatory constraints in the seafood industry (e.g., quotas, origin traceability). A transaction Req_i is considered valid if all validation rules are satisfied, see Eq. 8.

$$\forall R_k \in \mathcal{R}, \quad R_k(\text{Req}_i) = \text{true} \tag{8}$$

Non-compliance with any R_k results in rejection or legal issues. Therefore, regulatory compliance must be encoded as constraints within the SCs.

These mathematical formulations emphasize the importance of addressing on-chain cost, scalability, privacy, and compliance in real-world deployment scenarios of the proposed seafood requisition model.

5. Traceability via Ethereum Blockchain Let each seafood requisition transaction be modeled as a tuple, see Eq. 9.

$$\mathcal{T}_i = (ID_i, t_i, s_i, d_i, h(\text{data}_i)) \tag{9}$$

where:

- ID_i represents a unique transaction identifier,
- t_i is the timestamp of the transaction,
- s_i, d_i denote the sender and receiver addresses,
- $h(\text{data}_i)$ is the cryptographic hash of transaction metadata.

All $\mathcal{T}_i$ entries are recorded on ethereum's immutable ledger. Due to the collision-resistant property of $h(\cdot)$, transaction duplication or tampering becomes computationally infeasible. This ensures end-to-end traceability of seafood movements across the supply chain.

6. Fraud Prevention via zk-SNARKs Each requisition request is privately defined as shown in Eq. 10.

$$\text{Req}_i = \{q_i, c_i, s_i\} \tag{10}$$

where:

- q_i is the requested quantity,
- c_i represents a compliance or certification code,
- s_i is the supplier identity.

A zk-SNARKs proof π is generated for the constraint system $\mathcal{C}$ such that the requisition satisfies the condition in Eq. 11.

$$\mathcal{C}(\text{Req}_i) = 1 \Rightarrow \text{Valid requisition} \tag{11}$$

Rather than exposing sensitive data, the system submits $(\pi, h(\text{Req}_i))$, allowing a verifier to confirm validity through Eq. 12.

$$\text{Verify}(\pi, h(\text{Req}_i), \mathcal{C}) = \text{true} \tag{12}$$

This enables fraud detection without disclosing private requisition details, thereby ensuring data integrity and confidentiality.

7. Sustainability Compliance via SCs and zk-SNARKs Let L represent the maximum allowed quota (e.g., catch limit per cycle), and Q_u the quantity requested by user u. The sustainability constraint is expressed as Eq. 13:

$$Q_u \leq L \tag{13}$$

The system generates a zk-SNARKs proof π_L to validate this condition without revealing Q_u, see Eq. 14.

$$\text{Verify}(\pi_L, h(Q_u), Q_u \leq L) = \text{true} \tag{14}$$

SCs enforce this rule by rejecting any request where the condition in Eq. 15 is satisfied.

$$\text{Verify}(\pi_L, \cdot) = \text{false} \tag{15}$$

This mechanism ensures compliance with sustainability limits while preserving the users operational privacy.

The proposed framework leverages the ethereum blockchain and zk-SNARKs to address key challenges in the seafood requisition process. Ethereum provides a transparent and tamper-resistant ledger, while zk-SNARKs enable privacy-preserving verification without revealing sensitive requisition details. Together, these technologies form

a robust foundation for mitigating core industry concerns such as traceability, fraud prevention, sustainability enforcement, and regulatory compliance in blockchain-based seafood requisition systems. This section presents a structured analysis of how each concern is addressed through the integrated use of ethereum and zk-SNARKs.

However, While zk-SNARKs provide strong privacy guarantees by enabling verification without exposing sensitive requisition details, their integration introduces certain trade-offs. Generating zk-SNARKs proofs involves considerable computational overhead, which can impact the responsiveness of real-time seafood supply chains, especially when multiple requests are processed concurrently. Although the proofs are small and verify quickly, the setup and proof generation phases require optimized infrastructure to prevent delays. Additionally, scaling the framework to accommodate a large number of seafood harvesters and distributors increases the complexity of maintaining low-latency performance. These factors highlight the need for careful system design, including potential use of off-chain proof generation or batching strategies, to balance privacy preservation with operational efficiency.

8. Cost Considerations and Mitigation Strategies: One of the critical challenges in implementing blockchain and ZKPs in the seafood requisition system is the high initial deployment cost. This cost arises due to the infrastructure, computation, and integration requirements essential for system functionality. The total initial cost C_{init} can be formulated as shown in Eq. 16.

$$C_{\text{init}} = C_{\text{infra}} + C_{\text{ZKPs}} + C_{\text{dev}} + C_{\text{int}} \tag{16}$$

where:

- C_{infra}: Cost of setting up blockchain infrastructure (e.g., node hosting, network configuration),
- C_{ZKPs}: Computational cost of generating and verifying ZKPs (e.g., zk-SNARKs circuits),
- C_{dev}: Cost of SCs development and deployment,
- C_{int}: Integration cost with existing seafood supply chain systems.

These components collectively contribute to a high upfront financial requirement, particularly challenging for small-scale fisheries and distributors.

To address this, modular deployment strategies M_{mod} and shared infrastructure models I_{shared} can be adopted to distribute costs among stakeholders. Additionally, the integration of Layer-2 scaling protocols and off-chain ZKPs computation significantly reduces C_{ZKPs}, making the system more cost-efficient and accessible.

5.2 Key findings

The key findings of this study are summarized below to highlight the core contributions and outcomes of the proposed framework.

BT-CSRS Framework: This framework addresses challenges in seafood product return by leveraging ethereum and ZKPs to ensure secure, traceable, and privacy-preserving validation throughout the supply chain.

ZKPs-enabled Privacy Protection: ZKPs enable validation of sensitive information (e.g., origin, capture location, species) without disclosing the actual data, thereby preserving confidentiality while ensuring verifiability.

Traceability: The framework ensures that each stage of the seafood catching and handling process is verifiable and tamper-resistant, effectively reducing fraud and mislabeling.

Security Against Cyberattacks and Tampering: We have observed that the proposed framework enhances resistance to cyberattacks and data tampering by leveraging blockchain's immutable ledger and SCs automation, thereby strengthening overall security and ensuring reliable seafood transaction records.

Practical Implementation and Deployment: SCs, implemented using Solidity in remix IDE, operate on the ethereum sepolia test network, with metaMask serving as the blockchain wallet interface.

Performance Evaluation using JMeter: The proposed framework undergoes testing with the JMeter simulation toolkit, achieving the following results: *Throughput:* 682.36 TPS, *Latency:* 64.1 milliseconds, *Average Response Time:* 75.5 milliseconds, *STD Deviation:* 68.7 milliseconds

Comparative Superiority: We have observed that the proposed framework offers better performance and privacy-preserving capabilities compared to existing state-of-the-art seafood supply chain systems.

6 Conclusion

This study presents a comprehensive and privacy-preserving solution to the persistent challenges of the seafood supply chain through the proposed BT-CSRS framework, which integrates BT and ZKPs. By leveraging the ethereum blockchain, the system effectively addresses critical issues such as traceability, data confidentiality, fraud prevention, and sustainability compliance. BT-CSRS enables secure and verifiable seafood reacquisition and catching processes without disclosing sensitive information. SCs deployed on the ethereum sepolia test network enforce validation rules automatically, allowing only legitimate transactions while maintaining privacy through confidential ZKPs. These proofs validate details such as species, origin, and capture location without revealing actual data,

thereby ensuring both confidentiality and verifiability. The framework ensures end-to-end traceability by cryptographically recording each stage of seafood catching, handling, and return, significantly reducing risks of fraud, mislabeling, and unauthorized processing. The decentralized and immutable nature of blockchain, combined with automated SCs execution, strengthens system resilience against cyberattacks and tampering. SCs, developed using solidity in remix IDE and integrated through metaMask on the sepolia test network, support operational deployment. Performance testing through the JMeter simulation toolkit produced robust outcomes, with average throughput reaching 682.36 TPS, latency at 64.1 milliseconds, response time at 75.5 milliseconds, and a STD deviation of 68.7 milliseconds, confirming high responsiveness and scalability under diverse conditions. Additionally, BT-CSRS promotes environmentally sustainable practices by validating compliance with ethical sourcing and conservation standards, supporting responsible stakeholder decisions aligned with global sustainability efforts. The system further enhances transparency and trust by enabling consumers and regulators to verify the provenance and handling of seafood products across the supply chain. Comparative analysis indicates superior performance, enhanced privacy protection, and greater security in relation to existing frameworks. For future work, the framework can be extended to incorporate real-time IoT-based environmental sensors for dynamic compliance tracking, integrate multi-chain interoperability for broader supply chain collaboration, and explore advanced ZKPs systems such as zk-STARKs to further reduce computational overhead while maintaining privacy. These enhancements aim to strengthen the framework's scalability, efficiency, and adaptability for widespread adoption across global seafood ecosystems.

Author Contributions ● Pritam Rani: Study conception and design, Data collection, Analysis and interpretation of results, Draft manuscript preparation. ● Prity Rani: Study conception, analysis and interpretation of results, verification of results, draft manuscript preparation, and work supervision. ● Indrajeet Gupta: Supervised the work. ● Rohit Kumar Sachan: Supervised the work. ● Pratima Sharma: Supervised the work.

Funding No Funding of any sort if received for this manuscript.

Data Availability No datasets were generated or analysed during the current study.

Code Availability Not applicable to this article.

Declarations

Research Involving Human Participants and/or Animals Not applicable.

Conflicts of Interest Authors declare that no conflicts of interests of any sort is associated with the manuscript.

Informed Consent All the authors have confirmed the consent to submit the paper to Peer-to-Peer Networking and Applications.

Competing Interests The authors declare no competing interests.

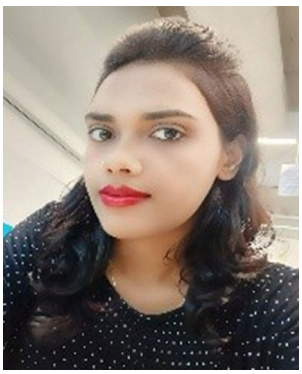
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